HERIOT WATT UNIVERSITY

SCHOOL OF SOCIAL SCIENCES

Edinburgh Business School

C21AO and C27IE

Economics / Introductory Economics

IN-PERSON EXAMINATION

SOLUTIONS

December Examination and Assessment Diet 2023

Duration: 2 hours

There are three sections to the examination paper
In Section A, there are **20 multiple choice questions**. You should attempt **all 20 questions**. Each question is worth 1.5 marks.
In Section B, there are **four** questions. You should attempt **all four questions**. Each question is worth 10 marks.
In Section C, there are **five** questions. You should attempt **any two questions**. Each question is worth 15 marks.

Attached: Multiple choice answer grid
Supplied: One sixteen page examination booklet
Calculator Permitted: Yes

Ethical Commitment: By submitting answers to this paper, I declare that these answers are entirely my own work and have not been shared, in part or in whole or in any form, with any other student, or disseminated in any way.

I also confirm that I have reviewed and understand all University regulations, policies and procedures on assessment misconduct, including but not limited to accessing unauthorised materials during this examination or the use of unauthorised electronic devices.

I understand that infringing this statement would represent an academic misconduct offence subject to disciplinary action according to the University Regulations and Policy regarding Academic Misconduct, with possible significant consequences for degree progression and final degree outcome.

MCQ Answer Grid

1	A
2	C
3	D
4	C
5	B
6	A
7	C
8	A
9	D
10	B
11	A
12	D
13	C
14	B
15	B
16	B
17	C
18	B
19	A
20	D

Section A

There are **20 multiple choice questions** in Section A. Please attempt **all these questions**, using the answer grid at the end of the examination, inserting the answer grid into your examination booklet when you are finished.

1) At the current market price of 120, there is excess demand for running shoes. We expect that:
   a) The market price will increase, the quantity supplied will increase, and the quantity demanded will decrease.
   b) The market price will decrease, the quantity supplied will increase, and the quantity demanded will decrease.
   c) The market price will decrease, the quantity supplied will decrease, and the quantity demanded will increase.
   d) The market price will increase, the quantity supplied will decrease, and the quantity demanded will increase.

2) Which of the following is not a problem to which we can easily apply microeconomic analysis?
   a) Understanding the response of firms to an increase in the wages which they must pay their workers.
   b) Understanding how the market price and firm output will change when new businesses enter a competitive market.
   c) Identifying a suitable government response to an energy price shock, which causes both unemployment and inflation.
   d) Estimating the margin over the competitive market price which a firm with market power can charge.

3) Which of the following are characteristics of the market for an abundant good?
   a) There will be excess supply of the good.
   b) The good will be freely available to all potential users of it.
   c) People will not face any limits on their use of the good.
   d) All three statements in parts a) – c).

4) Coca Cola produces many sugary drinks, including Sprite and Fanta. Assuming that there is limited substitutability between Sprite and Fanta, we expect that the cross price elasticity of demand will be:
   a) Less than -1.
   b) Between -1 and 0.
   c) Between 0 and 1.
   d) More than 1.

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5) A fashionable cookery writer recommends using mangos in dessert in a viral Tiktok at the same time as trade unions negotiate higher wages for workers on mango farms. We expect that:
   a) The demand curve and the supply curve will both shift to the left.
   b) The demand curve will shift to the right and the supply curve will shift to the left.
   c) The demand curve will shift to the left and the supply curve will shift to the right.
   d) The demand curve and the supply curve will both shift to the right.

6) In many developing countries, cars are superior goods. In these countries:
   a) The income elasticity of demand for cars is greater than one.
   b) The income elasticity of demand is less than zero.
   c) The price elasticity of cars is less than minus one.
   d) The price elasticity of cars is greater than zero.

7) Roots and Fruits operates in a monopolistically competitive market. Analysing its costs of production, we expect to find that:
   a) Its average variable costs are greater than its average costs.
   b) Its average fixed costs increase when they are less than its marginal costs.
   c) Its average variable costs are at their minimum when they are equal to its marginal costs.
   d) Its marginal costs continue to fall as it expands its output.

8) KTM is the main train operator in Malaysia, and so has substantial monopoly power. As a result, we expect that:
   a) It will restrict services to increase its profits.
   b) It will not be able to generate long-run economic profits.
   c) It will adopt marginal cost pricing to maximise its profits.
   d) It will set lower train fares than train operators in competition with each other.

9) Driving into work, when traffic slows down on the motorway because there is traffic congestion, this is evidence of:
   a) The road network being a public good.
   b) The government regulating an externality efficiently.
   c) Driving on the motorway being a non-rivalrous activity.
   d) Access to the road system being non-excludable.

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10) The market for mobile phones is dominated by a small number of large businesses, so has the form of an oligopoly. We expect that:
   a) These firms need to accept the market price.
   b) These firms may be able to collude to raise the market price.
   c) The market price will be lower than if the market were to become more competitive.
   d) The market price will ensure that firms only make normal profits, and break even.

11) After the government imposes a tariff on imports of steel, which group can we be certain will be worse off?
   a) Firms which use steel in production.
   b) The Government.
   c) Domestic firms which produce steel.
   d) Workers employed in the steel industry.

12) If the Malaysian ringgit appreciates against the pound sterling, then:
   a) Malaysian students in Edinburgh will find that the cost of their tuition fees (in ringgits) has increased.
   b) When the University makes payments from Malaysia to Edinburgh, it will have to pay more (in ringgits).
   c) There will be inflationary pressures in Malaysia.
   d) Malaysia will find that demand for its goods in the UK has fallen.

13) Which of the following is not a basis for calculating the GDP measure of national income?
   a) The expenditure on all final goods and services produced within the economy.
   b) The incomes received by households from the provision of factors of production.
   c) The transfers which the government makes to support disadvantaged families.
   d) The value added in the production of goods and services.

14) Why might the multiplier effect on income of a reduction in taxation be less than the multiplier effect of an increase in government expenditure?
   a) Government can only spend money on domestic goods and services.
   b) Households are likely to save some of the reduction in taxation.
   c) Businesses charge the government higher prices than households.
   d) There will be more investment after an increase in government spending.

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15) Which statement best captures the extreme Keynesian beliefs about the shape of the aggregate supply curve?
   a) The long run aggregate supply curve is upward sloping, but steeper than the short run aggregate supply curve.
   b) The long run aggregate supply curve is horizontal.
   c) The short run aggregate supply curve is horizontal up to the capacity of the economy, and then becomes vertical.
   d) The short run aggregate supply curve is vertical.

16) When the central bank sells short term government bills to the commercial banks:
   a) The government is reducing its use of deficit financing.
   b) The central bank is reducing the money supply.
   c) The central bank is using monetary policy to reduce interest rates.
   d) The government is running an expansionary monetary policy.

17) If taxation is a quarter of income, there is no saving, and one third of disposable income (income after tax has been paid) is spent on imports, then the marginal propensity to spend on domestic goods is:
   a) 0.75
   b) 0.67
   c) 0.5
   d) 0.33

18) Which of the following is a supply side policy?
   a) Reducing the demand for credit by increasing the discount rate of interest on lending by the central bank to commercial banks.
   b) Reducing regulations associated with hiring workers and terminating their contracts.
   c) Increasing government spending, with the deficit funded by the issue of long-term bonds.
   d) Increasing the money supply with the central bank purchasing government bills held by commercial banks.

19) Which of following is a statement of the natural rate hypothesis?
   a) There is level of unemployment at which the rate of inflation will remain static.
   b) There is a level of national income at which there will be full employment.
   c) There is a level of growth of national income which is consistent with macroeconomic stability.
   d) There is a level of the growth of the money supply which ensures that there will be persistent economic growth.

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20) When there is an increase in investment, increasing the productive capacity of the economy, we might capture the effects in an aggregate demand-aggregate supply diagram with:
   a) A shift of the aggregate demand curve to the left
   b) A shift downwards of the short-run aggregate supply, while the long-run aggregate supply curve does not move.
   c) A downward shift of the aggregate demand curve, while the long-run aggregate supply curve becomes flatter.
   d) A shift to the right of both the aggregate demand curve and the long-run aggregate supply curve.

[Total: 30 marks]

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Section B

There are **four short answer questions** in this section, all of which require you to carry out some calculations. You should answer **all four questions**. Each question is worth 10 marks.

1) Crystal Clear Audio makes headsets which can be used with laptop computers. It has estimated demand for its output at prices from 10 up to 60.

Price	0	10	20	30	40	50	60
Output (thousands)	-	3,600	1,800	1,200	900	720	600

a) Define the concept of the price elasticity of demand, and explain how to measure it. [2 marks]

Looking for responsive of demand to price change and the ratio of either proportional or percentage changes in output and price.

b) Using the mid-point formula for the price elasticity of demand, calculate the price elasticity of demand for headsets when the price changes from 10 to 20, and also when it changes from 10 to 60. [4 marks]

Expect formula PED = $\frac{\frac{Q_1 - Q_0}{Q_1 + Q_0}}{\frac{p_1 - p_0}{p_1 + p_0}}$; so when between prices 10 and 20,

PED = $\frac{\frac{1800 - 3600}{3600 + 1800}}{\frac{20 - 10}{20 + 10}} = \frac{-\frac{1}{3}}{\frac{1}{3}}$ = -1, while between prices 10 and 60,

PED = $\frac{\frac{600 - 3600}{3600 + 600}}{\frac{60 - 10}{60 + 10}} = \frac{-\frac{5}{7}}{\frac{5}{7}}$ = -1

c) Comment on the nature of the price elasticity of demand, and how demand changes as price changes. Without doing any further calculations, use the value of the price elasticity of demand to state what will happen to Crystal Clear Audio's sales revenue as it increases its prices, and the value of its marginal revenue. [4 marks]

Since demand has unit price elasticity, then the change in demand will be equiproportionate with the change in price, but in the opposite direction, so that as price increases, demand falls. The total revenue will be constant. With constant total revenue, marginal revenue is zero.

[Total: 10 marks]

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2) Crystal Clear Audio now diversifies into the production of video cameras. It has estimated how the number of cameras which it could produce in a year will change with the number of workers which it hires.

a) Define the average physical product and the marginal physical product. Complete the table below. [4 marks]

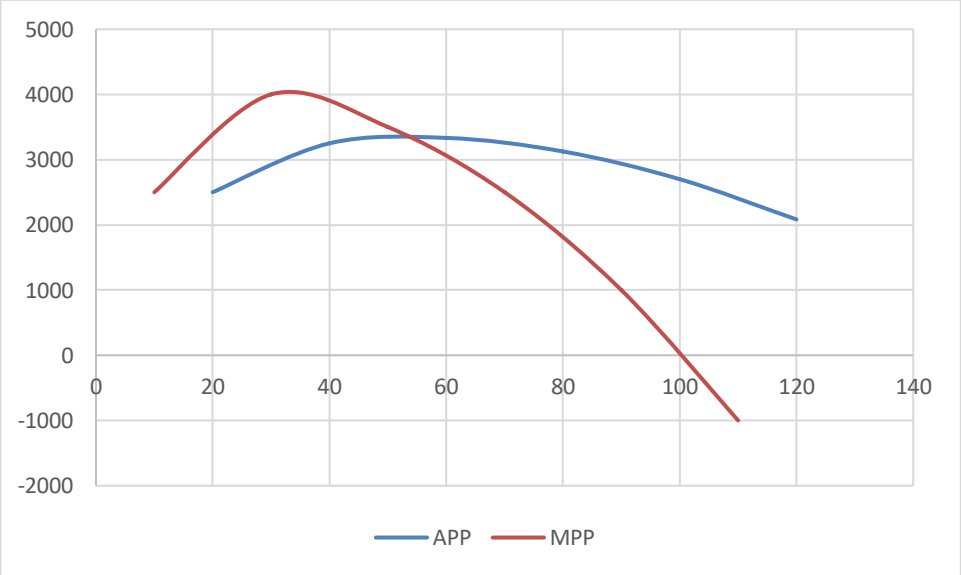
Average physical product is the number of cameras produced per worker. Divide output by the number of workers.

Marginal physical product is the additional number of cameras which Crystal Clear Audio could produce by hiring one more worker. Divide the change in output from hiring 20 more workers by 20

Workers	0	20	40	60	80	100	120
Output (thousands)	0	50	130	200	250	270	250
APP	-	2,500	3,250	3,333	3,125	2,700	2,083
MPP		2,500	4,000	3,500	2,500	1,000	-1,000

b) Sketch a diagram showing the average physical product and the marginal physical product. In your sketch, show clearly where the average physical product and the marginal physical product are equal. [4 marks]

In a diagram with physical product measured on the vertical axis and workers on the horizontal axis, both curves have maximum points. I expect to see the maximum of the MPP curve between 20 and 40 workers, and the maximum of the APP curve between 40 and 60 workers, with the MPP curve passing through maximum.



c) Assume that Crystal Clear Audio pays its workers 30,000. Given that wages are the only variable costs, complete the table below, showing the average variable cost of output. [2 marks]

Output (thousands)	0	50	130	200	250	270	250
TVC (thousands)	0	600	1,200	1,800	2,400	3,000	3,600
AVC	-	12	9.23	9	9.6	11.11	14.40

We require TVC = 30,000W, and AVC = TVC/Q

[Total: 10 marks]

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3) You are to analyse this model of aggregate demand.
For national income (GDP) *Y,* aggregate consumption, *C*, savings, *S*, investment, *I*, government expenditure, *G*, taxation *T*, income after taxation, Y^D, exports, *X* and imports, *M*, where

$C = 0.8Y^D$
$T = 0.25Y$
$M = 0.2Y$
$G = 1{,}000$, $I = 500$, $X = 900$

a) State an equilibrium condition, give an economic interpretation of it, and then use it to find the equilibrium level of national income, Y^* [5 marks]

Students have a choice of $Y = C + I + G + X - M$ *(income equals expenditure within the economy) or* $S + T + M = I + G + X$ *(withdrawals equals injections).* Then in equilibrium, $Y^* = 0.8*0.75Y^* + 500 + 1{,}000 + 900 - 0.2Y^*$, so that $0.6Y^* = 2{,}400$, and $Y^* = 4{,}000$.

b) Define the government expenditure multiplier, explain why it typically has a value which is greater than one, but finite, and calculate its value in this case. [5 marks]

Government expenditure multiplier is the ratio of the increase in national income to an increase in national income. It typically takes a value greater than one because spending by government increases factor incomes, which leads to increases in consumption and investment. However, since there are leakages from the circular flow through savings, taxation and the import of goods, these effects will decay.
The multiplier $M = (MPW)^{-1}$. In this case, 25% of income is paid in taxes, 15% is saved, and 20% is spent on imports. So 60% of income is income is withdrawn, giving a multiplier value of $M = (0.6)^{-1} = 5/3$.

4) Atlantis and Oceania produce two fruits, melons and figs. Each has a workforce of 25,000 people. In the table below, you can see the quantities which each country would produce were the whole of its workforce engaged in the production of each fruit. You may assume that as workers shift between production of figs and melons, the marginal product of both industries remains constant.

Capacity (thousand tonnes)	Country	
	Atlantis	Oceania
Melons	200	300
Figs	50	60

Without trade, both countries choose to produce a combination of both fruits. Atlantis produces 120,000 tonnes of melons and 20,000 tonnes of figs; and Oceania produces 150,000 tonnes of melons, and 30,000 tonnes of figs.

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a) Define the concept of *absolute advantage*. State which country has the absolute advantage in the production of each fruit. [2 marks]

The country with the higher labour productivity in production of a good has an absolute advantage. In this case, Oceania has an absolute advantage in the production of both goods.

b) Define the concept of comparative advantage. Explain why it is impossible for either country to have a comparative advantage in the production of both goods, and state for the production of which good each country has a comparative advantage. [3 marks]

Comparative advantage occurs in the country which has the lower opportunity cost of production. Since the opportunity cost of producing melons is the inverse of the opportunity cost of producing figs, neither country can have the comparative advantage in both goods. With the opportunity cost of a tonne of figs equal to four tonnes of melons in Atlantis, and five tonnes of melons in Oceania, Atlantis has the comparative advantage in figs, and Oceania has the comparative advantage in producing melons.

c) The fig farmers of Atlantis have claimed that it is possible that if the country gives up melon production, and trades with Oceania, it will be possible to consume more figs and more melons. Examine this claim. [5 marks]

Given that Atlantis is currently producing 20,000 tonnes of figs and 120,000 tonnes of melons, by specialising in the production of figs, it could increase output to 50,000 tonnes. Assuming that 30,000 tonnes of figs were then sold to Oceania, it would be possible for farmers in Oceania to increase production of melons to up to 300,000 tonnes. For the farmers' calculations to be correct, farmers in Oceania must continue to produce some figs, but melon production cannot increase unless their output falls below 6,000 tonnes.

[Total: 10 marks]

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Section C

Discussion questions

1) Complete a PEST analysis of the market environment facing an international supplier of higher education, which is considering setting up business in a new country. [15 marks]

   Students should define political, economic, social, and technological considerations. Beginning with technological challenges, they should be aware of developments in higher education, including the use of virtual learning environments, and content sharing between faculty which will allow for the rapid development of standardised teaching in the new country. They may also be aware of technological developments such as the emergence of AI, which will enable knowledge to be collated much more efficiently.

   Social considerations should include how the concept of education which the provider has developed will align with the traditions of education in the country in which it intends to operate.

   Political considerations will include the likelihood that higher education will be subject to licensing and regulatory requirements, intended to ensure that students are offered high quality services.

   Economic considerations will relate to the extent to which the organisation will be able to manage costs, for example by hiring local, rather than international, staff. They will also need to consider the fee structure which they will be able to adopt, and be aware of the extent of competition within the market, and the extent to which competition occurs in prices, or quality – the latter having implications for marketing expenditure.

2) In many countries, we find that there is dominant supplier of energy for domestic use, which has substantial market power.

   a) Applying the Structure-Conduct-Performance approach, explain the likely differences between a monopoly and a competitive industry. [6 marks]
   Students should concentrate on the fact that a monopolist tends to be certain that it will not face substantial entry, offering explanations which are based on increasing returns to scale in production, which give the monopolist a cost advantage, or through the ability to prevent entry, for example through the ownership of essential inputs, or through intellectual property such as patent rights.

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b) Using a diagram, explain why we expect that a profit-maximising monopoly will produce less output, and charge a higher price than if it were to produce the socially optimal output. [6 marks]

In a diagram with a downward sloping demand (average revenue) curve, and the marginal revenue curve below it, there will also be an average cost curve, which will slope upwards only at high values of outputs, with the upward sloping marginal cost curve then cutting through it. Students should claim that profit maximisation occurs when MC = MR (so that increasing output, costs exceed revenues), while the social optimum occurs where the price (or marginal benefits) equal marginal cost (so that increasing outputs, benefits from consumption are less than costs of generating them). With MR curve below the demand curve, the result follows.

c) Briefly explain why we may prefer to have a monopoly supplier than insist upon competition in markets. [3 marks]

This requires a short note explaining that while competition tends to lead to the social optimum, it will not do so in markets in which there are substantial economies of scale; or where there is good reason to expect that larger organisations will be able to lower costs of production more quickly than small ones.

[Total: 15 marks]

3) In economic analysis, we consider climate change to be one of the most important examples of a negative externality. Using a diagram if you consider that to be sensible, explain how we analyse the effects of an externality, and set out how this approach might be applied to the case of climate change. [15 marks]

Students will probably use a graph to demonstrate the analysis of the economically efficient level of a negative production externality. On a diagram with the level of an activity measured on the horizontal axis, and (monetary) values measured on the vertical axis, student will distinguish between the marginal private costs of production, taken into account by firms in setting output, and the higher marginal social costs, which reflect the costs of production to society. Drawing a marginal benefits curve, students should note that the social optimum typically occurs when there is a lower level of output, which takes place at a higher price.

Turning to the application to climate change, good answers will note that the emission of greenhouse gases is typically a by-product of other activities, so that changing the nature of those activities so that they lead to lower emissions is the general objective.

Students might then note the variety of measures which can be taken to manage an externality, including quota, the imposition of taxes, regulation, and bans, noting that in the case of climate change there are particular problems of global coordination because the places which feel the effects of climate change – and which will feel them in the future – often have contributed little to the emission of greenhouse gases.

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4) For many years, economists have used the term ‘business cycle’ to refer to the succession of periods of economic expansion, followed by recession, over approximately 7 – 9 years. An important objective of government economic policy is to manage these fluctuations.

a) Briefly outline the links between the business cycle and the credit cycle in banking. [6 marks]

Students should note that the cycle is often associated with changes in the availability of credit, which in turn reflect confidence in the future path of the economy. Banks make investment funds available at times when economic prospects seem positive, and they are confident of borrowers repaying. However, suffering losses from loan failures, banks need to rebuild their balance sheets, and raise more capital. While doing that, they reduce lending, so that investment declines rapidly, and economic growth declines. This can cause a loss of confidence in future economic prospects, leading to a decline in the demand for investment funds.

b) How might governments design their policies so that measures to reduce the fluctuations in the business cycle either occur automatically or else follow rules? [9 marks]

Students should be able to define automatic stabilisers as contingent spending commitments which increase when the economy is weak. Unemployment benefits naturally increase during recessions because of an increase in the number of claims. They should also know that fiscal rules commit the government to varying expenditure as the state of the economy changes.

They should be aware that the alternative to the use of rules is the use of discretion. Giving credit for noting that one reason for democratic governments choosing to follow rules is the risk of increasing volatility though the interaction of the electoral cycle with the business cycle, here I expect students to concentrate on the difficulties of designing a discretionary policy – the identification lag, the design and approval lags, and the implementation lag, so that a policy might only be effective a year or so after the economic conditions occur which justify it, by which time the state of the economy is likely to have changed. Very good answers will note that there is also an argument that the private sector of the economy is more likely to make long-term commitments where the economic environment is predictable, so that discretionary policy may reduce the effectiveness of supply-side policies.

[Total: 15 marks]

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5) The exchange rate between the US dollar (USD) and the Euro (EUR) changes every day. We define it as the number of Euros required to buy one US dollar.

a) List the four sources of variability in the exchange rate, briefly stating their effect. [8 marks]

Students should think of the exchange rate as the price of the dollar in terms of Euros. The dollar becomes more expensive if demand for it increases, or supply decreases. The exchange rate should therefore increase (and the dollar will appreciate) if

- US net exports increase, or Eurozone net exports decrease
- Either the flow of (financial) investment into the USA increases, or else it declines in the Eurozone
- Inflation falls in the USA relative to the Eurozone, so that the purchasing power of the US dollar increases relative to the Euro
- The value of the dollar is expected to appreciate, so that traders purchase in anticipation of increased demand – with the expectations becoming self-fulfilling.

b) Discuss how exchange rate variability might affect the government's ability to support economic growth. [7 marks]

While specific effects depend on effects within economic sectors, expect students to note that exchange rate variability tends to reduce the effectiveness of demand management policies. For example, an increase in government spending, funded by borrowing, will tend to increase the interest rate, causing a flow of investment funds into the country, so that the currency appreciates. Additional spending will tend to increase imports while the currency appreciation will tend to depress exports. Net exports will therefore decline, reducing aggregate demand. Students may also choose examples involving the management of interest rates, and take into account the inflationary impact of increased demand.

[Total: 15 marks]

END OF PAPER

On the following page, you will find the multiple choice answer grid.

Section

Answer Grid for the multiple choice questions

Please place a cross in a box to indicate clearly your chosen response to each question. If you wish to change your response, score through the box which you had chosen first, and then place a cross in the box corresponding to your preferred response.

	Answer			
Question	A	B	C	D
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